mischievously described as “highways.” We think we (barely) understand these landforms in terms of faults and collisions, but of course we might be wrong.

The surface stains of organic matter—sometimes, as on Triton, delicately hued—are attributed to charged particles producing chemical reactions in simple hydrocarbon ices, generating more complex organic materials, and all this having nothing to do with the intermediation of life. But of course we might be wrong.

The complex pattern of radio static, bursts, and whistles that we receive from all four Jovian planets seems, in a general way, explicable by plasma physics and thermal emission. (Much of the detail is not yet well understood.) But of course we might be wrong.

We have found nothing on dozens of worlds so clear and striking as the signs of life found by the *Galileo* spacecraft in its passages by the Earth. Life is a hypothesis of last resort. You invoke it only when there’s no other way to explain what you see. If I had to judge, I would say that there’s no life on any of the worlds we’ve studied, except of course our own. But I might be wrong, and, right or wrong, my judgment is necessarily confined to this Solar System. Perhaps on some new mission we’ll find something different, something striking, something wholly inexplicable with the ordinary tools of planetary science—and tremulously, cautiously, we will inch toward a biological explanation. However, for now nothing requires that we go down such a path. So far, the only life in the Solar System is that which comes from Earth. In the Uranus and Neptune systems, the only sign of life has been *Voyager* itself.

As we identify the planets of other stars, as we find other worlds of roughly the size and mass of the Earth, we will scrutinize *them* for life. A dense oxygen atmosphere may be detectable even on a world we’ve never imaged. As for the Earth, that may by itself be a sign of life. An oxygen atmosphere with appreciable quantities of methane would almost certainly be a sign of life, as would modulated radio emission. Someday, from observations of our planetary system or another, the news of life elsewhere may be announced over the morning coffee.